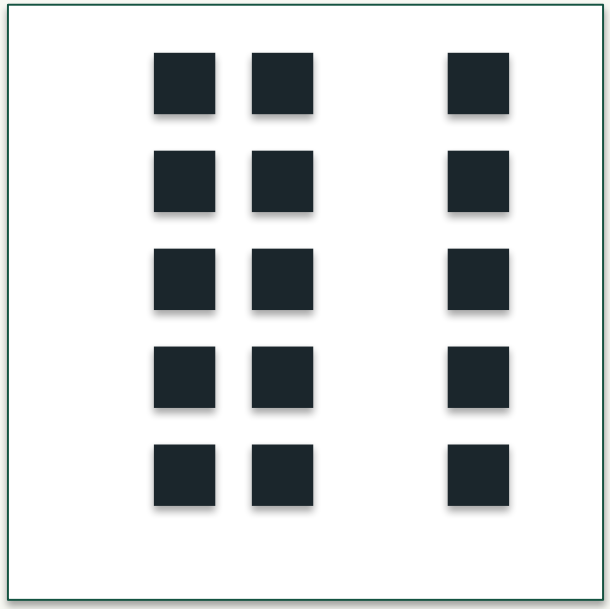
Sustainability Reporting and Engineering Strategies for Senior Leaders



TGS-2025052343
Version 5.0 • 01 Oct 2026

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Digital Attendance



✓ Successful check-in

Verify learner name, course run and session. If the success state is absent, notify the trainer before proceeding.

ADMIN

Trainer profile template

WHAT

? avatar • verified qualifications • relevant experience • adult-education credentials

VERIFY

Confirm the displayed course, run and learner record.

OWNER

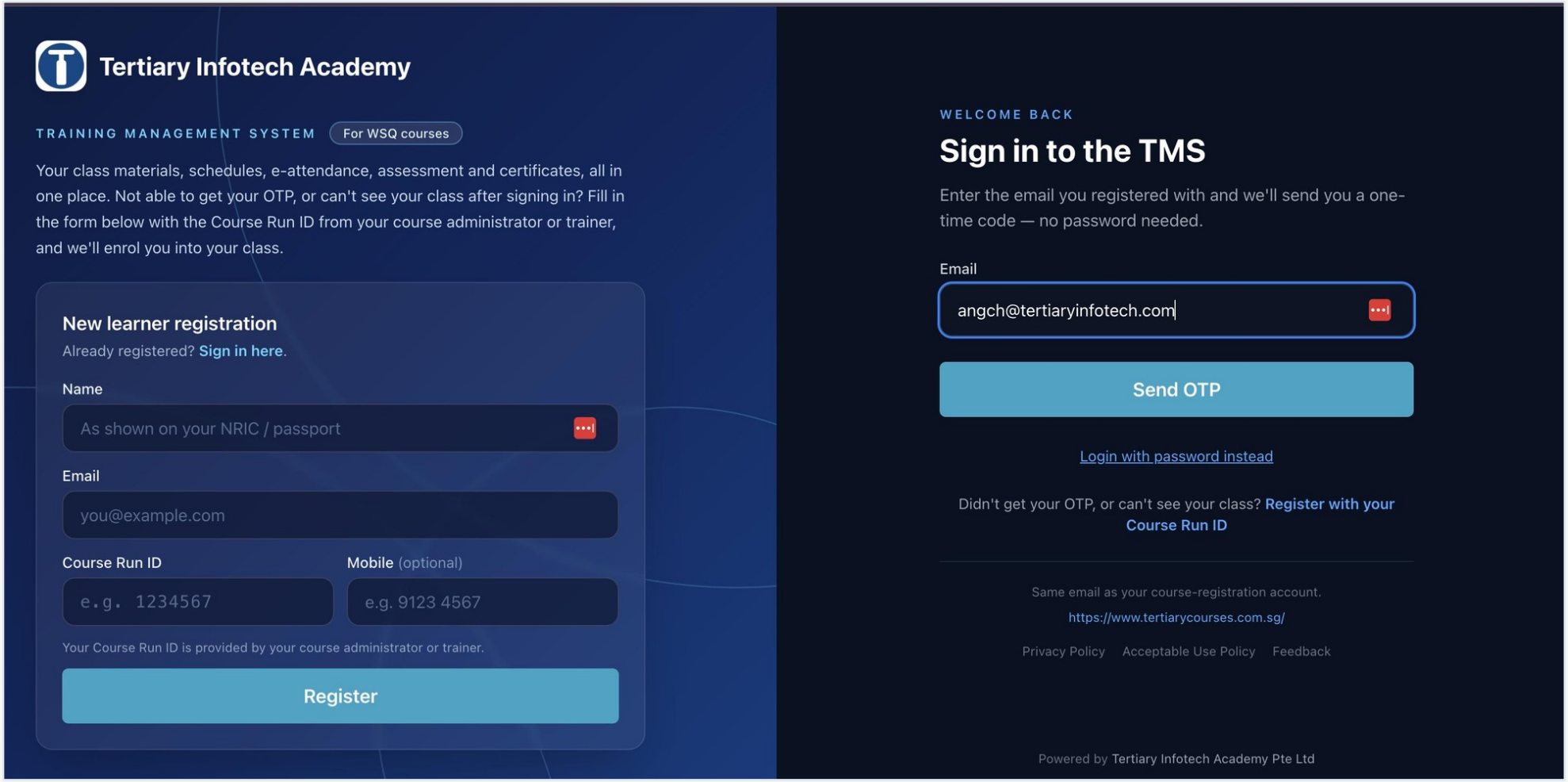
Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR TRAINER EVIDENCE FIELDS

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

ADMIN

Dr Alfred Ang

WHAT

Named trainer • profile details require verified records

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR NAMED PROFILE PLACEHOLDER

Learning outcomes

WHAT

A1 endorse • A3 benchmark • A2 comply • A4 propose

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR K/A OUTCOME MAP

Two-day journey

WHAT

Day 1: LU1–2 • Day 2: LU3–4 • WA + CS

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ADMIN

Assessment briefing

WHAT

WA-SAQ 60 min • Case Study 60 min • open book • individual

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR ASSESSMENT CONDITIONS

ADMIN

Assessment evidence

WHAT

Five WA questions K1–K5 • four CS tasks A1/A3/A2/A4

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR ASSESSMENT COVERAGE

ADMIN

Skills Framework

WHAT

Engineering Services • Sustainable Engineering • EGS-EPM-6065-1.1

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR TSC IDENTITY

ADMIN

Activity roadmap

WHAT

Option appraisal • materiality • controls • decarbonisation

VERIFY

Confirm the displayed course, run and learner record.

OWNER

Trainer / assessor / course administrator

RECORD

Retain the approved attendance or assessment record.

ANCHOR FOUR-ACTIVITY TRACE

Engineering research currency: operating model

INPUT

Research register

TRANSFORM

TRL–evidence matrix

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR TRL–EVIDENCE MATRIX

Engineering research currency: worked calculation

FORMULA / DECISION LOGIC

Evidence score = recency × relevance × reproducibility

SYNTHETIC TEACHING EXAMPLE

6 current studies × 0.9 relevance × 0.8 reproducibility = 4.32 weighted evidence units; hold endorsement below 4.0.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Endorse only with verified operating evidence Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR EVIDENCE SCORE = RECENCY × RELEVANCE × REPRODUCIBILITY

Engineering research currency: evidence artifact

ACTUAL ARTIFACT

Research register

REQUIRED FIELDS

Engineering research currency boundary; reporting period; Evidence score = recency × relevance × reproducibility; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Outdated pilot used as proof The artifact supports only the stated boundary and period.

ANCHOR RESEARCH REGISTER

Engineering research currency: failure diagnostic

FAILURE

Outdated pilot used as proof

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Endorse only with verified operating evidence

ANCHOR OUTDATED PILOT USED AS PROOF

Engineering research currency: executive decision gate

REQUIRED INPUT

Research register

MEASUREMENT

Evidence score = recency × relevance × reproducibility Worked basis: 6 current studies × 0.9 relevance × 0.8 reproducibility = 4.32 weighted evidence units; hold endorsement below 4.0.

TOPIC-SPECIFIC THRESHOLD

Endorse only with verified operating evidence

RETAINED RECORD

Engineering research currency decision record: source, calculation, reviewer, conditions and next review.

ANCHOR ENDORSE ONLY WITH VERIFIED OPERATING EVIDENCE

Sustainability system boundary: operating model

INPUT

Mass-balance worksheet

TRANSFORM

Input–process–output boundary

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR INPUT–PROCESS–OUTPUT BOUNDARY

Sustainability system boundary: worked calculation

FORMULA / DECISION LOGIC

Balance variance = inputs − outputs − stock change

SYNTHETIC TEACHING EXAMPLE

1,000 t input − 720 t product − 210 t recovered − 55 t waste = 15 t unexplained variance (1.5%); investigate above 1%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Freeze boundary before option comparison Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR BALANCE VARIANCE = INPUTS − OUTPUTS − STOCK CHANGE

Sustainability system boundary: evidence artifact

ACTUAL ARTIFACT

Mass-balance worksheet

REQUIRED FIELDS

Sustainability system boundary boundary; reporting period; Balance variance = inputs − outputs − stock change; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Scope shifted to improve result The artifact supports only the stated boundary and period.

ANCHOR MASS-BALANCE WORKSHEET

Sustainability system boundary: failure diagnostic

FAILURE

Scope shifted to improve result

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Freeze boundary before option comparison

ANCHOR SCOPE SHIFTED TO IMPROVE RESULT

Sustainability system boundary: executive decision gate

REQUIRED INPUT

Mass-balance worksheet

MEASUREMENT

Balance variance = inputs − outputs − stock change
Worked basis: 1,000 t input − 720 t product − 210 t recovered − 55 t waste = 15 t unexplained variance (1.5%); investigate above 1%.

TOPIC-SPECIFIC THRESHOLD

Freeze boundary before option comparison

RETAINED RECORD

Sustainability system boundary decision record: source, calculation, reviewer, conditions and next review.

ANCHOR FREEZE BOUNDARY BEFORE OPTION COMPARISON

Public policy horizon: operating model

INPUT

Policy obligations register

TRANSFORM

Policy-to-design trace

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR POLICY-TO-DESIGN TRACE

Public policy horizon: worked calculation

FORMULA / DECISION LOGIC

Compliance lead time = effective date − design freeze

SYNTHETIC TEACHING EXAMPLE

Obligation starts 1 Jan 2028; design freeze 1 Jul 2026 → 18-month lead time; trigger specification update now.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Label enacted, proposed and voluntary separately Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR COMPLIANCE LEAD TIME = EFFECTIVE DATE − DESIGN FREEZE

Public policy horizon: evidence artifact

ACTUAL ARTIFACT

Policy obligations register

REQUIRED FIELDS

Public policy horizon boundary; reporting period; Compliance lead time = effective date − design freeze; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Consultation treated as law The artifact supports only the stated boundary and period.

ANCHOR POLICY OBLIGATIONS REGISTER

Public policy horizon: failure diagnostic

FAILURE

Consultation treated as law

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Label enacted, proposed and voluntary separately

ANCHOR CONSULTATION TREATED AS LAW

Public policy horizon: executive decision gate

REQUIRED INPUT

Policy obligations register

MEASUREMENT

Compliance lead time = effective date − design freeze Worked basis: Obligation starts 1 Jan 2028; design freeze 1 Jul 2026 → 18-month lead time; trigger specification update now.

TOPIC-SPECIFIC THRESHOLD

Label enacted, proposed and voluntary separately

RETAINED RECORD

Public policy horizon decision record: source, calculation, reviewer, conditions and next review.

ANCHOR LABEL ENACTED, PROPOSED AND VOLUNTARY SEPARATELY

Energy management: operating model

INPUT

Meter-to-KPI evidence chain

TRANSFORM

Energy baseline model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR ENERGY BASELINE MODEL

Energy management: worked calculation

FORMULA / DECISION LOGIC

Intensity = kWh ÷ production unit

SYNTHETIC TEACHING EXAMPLE

120,000 kWh ÷ 10,000 units = 12.0 kWh/unit versus 13.5 baseline → 11.1% improvement.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Normalise before claiming improvement Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR INTENSITY = KWH ÷ PRODUCTION UNIT

Energy management: evidence artifact

ACTUAL ARTIFACT

Meter-to-KPI evidence chain

REQUIRED FIELDS

Energy management boundary; reporting period; Intensity = kWh ÷ production unit; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Weather/volume effect ignored The artifact supports only the stated boundary and period.

ANCHOR METER-TO-KPI EVIDENCE CHAIN

Energy management: failure diagnostic

FAILURE

Weather/volume effect ignored

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Normalise before claiming improvement

ANCHOR WEATHER/VOLUME EFFECT IGNORED

LU1 retained source evidence: annotated decision use



SOURCE

Original slide 22 from “WSQ - Master Trainer Slides - Sustainability Reporting and Engineering Strategies for Senior Leaders - v3.pptx”.

INTERPRET

Inspect the depicted sustainability mechanism; name its system boundary, operating evidence and endorsement condition.

CONTROL

Record boundary, method, owner, threshold and source citation before using the evidence.

Water stewardship: operating model

INPUT

Water-flow diagram

TRANSFORM

Site water balance

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR SITE WATER BALANCE

Water stewardship: worked calculation

FORMULA / DECISION LOGIC

Reuse rate = reused water ÷ total demand

SYNTHETIC TEACHING EXAMPLE

48,000 m³ reused ÷ 120,000 m³ demand = 40% reuse; target 45% → 5 percentage-point gap.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Approve only after catchment context review Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR REUSE RATE = REUSED WATER ÷ TOTAL DEMAND

Water stewardship: evidence artifact

ACTUAL ARTIFACT

Water-flow diagram

REQUIRED FIELDS

Water stewardship boundary; reporting period; Reuse rate = reused water ÷ total demand; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Withdrawal confused with consumption The artifact supports only the stated boundary and period.

ANCHOR WATER-FLOW DIAGRAM

Water stewardship: failure diagnostic

FAILURE

Withdrawal confused with consumption

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

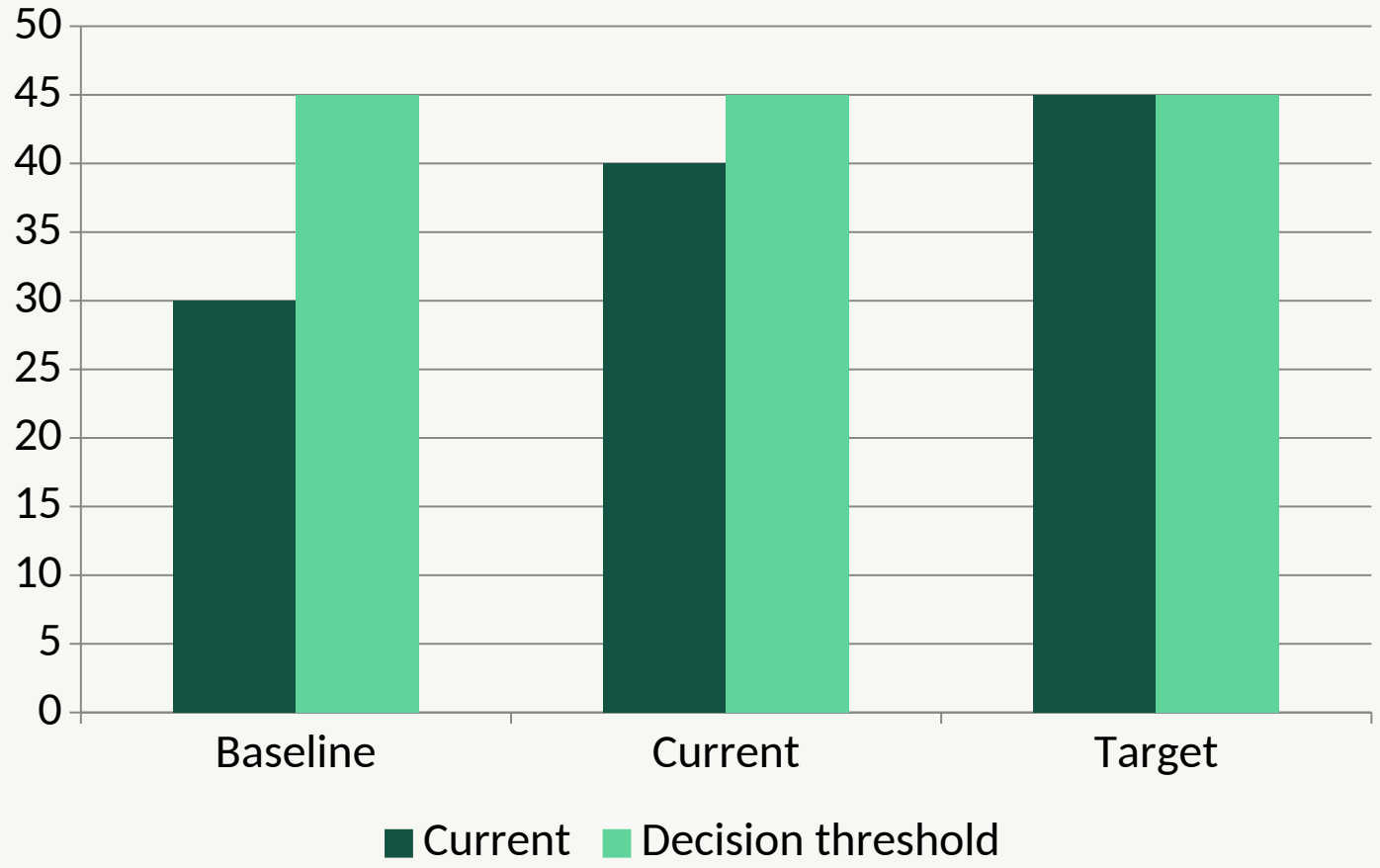
Reperform the calculation and trace the source.

REMEDIATE

Approve only after catchment context review

ANCHOR WITHDRAWAL CONFUSED WITH CONSUMPTION

Water stewardship: executive decision gate



REQUIRED INPUT

Water-flow diagram

MEASUREMENT

Reuse rate = reused water ÷ total demand Worked basis: 48,000 m³ reused ÷ 120,000 m³ demand = 40% reuse; target 45% → 5 percentage-point gap.

ANCHOR APPROVE ONLY AFTER CATCHMENT CONTEXT REVIEW

Engineering option appraisal: operating model

INPUT

CAPEX/payback/impact table

TRANSFORM

Weighted option matrix

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR WEIGHTED OPTION MATRIX

Engineering option appraisal: worked calculation

FORMULA / DECISION LOGIC

Score = Σ rating × weight

SYNTHETIC TEACHING EXAMPLE

(4×25%)+(5×25%)+(3×15%)+(4×20%)+(4×15%) = 4.10/5; approve above 3.5.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Approve weights before scoring Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR SCORE = Σ RATING × WEIGHT

Engineering option appraisal: evidence artifact

ACTUAL ARTIFACT

CAPEX/payback/impact table

REQUIRED FIELDS

Engineering option appraisal boundary; reporting period; Score = Σ rating × weight; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Weights chosen after seeing result The artifact supports only the stated boundary and period.

ANCHOR CAPEX/PAYBACK/IMPACT TABLE

Engineering option appraisal: failure diagnostic

FAILURE

Weights chosen after seeing result

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Approve weights before scoring

ANCHOR WEIGHTS CHOSEN AFTER SEEING RESULT

Engineering option appraisal: executive decision gate

REQUIRED INPUT

CAPEX/payback/impact table

MEASUREMENT

Score = Σ rating × weight Worked basis: (4×25%) +(5×25%)+(3×15%)+(4×20%)+(4×15%) = 4.10/5; approve above 3.5.

TOPIC-SPECIFIC THRESHOLD

Approve weights before scoring

RETAINED RECORD

Engineering option appraisal decision record: source, calculation, reviewer, conditions and next review.

ANCHOR APPROVE WEIGHTS BEFORE SCORING

Life-cycle thinking: operating model

INPUT

Cradle-to-gate inventory

TRANSFORM

Life-cycle stage map

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR LIFE-CYCLE STAGE MAP

Life-cycle thinking: worked calculation

FORMULA / DECISION LOGIC

Impact = Σ activity data × factor

SYNTHETIC TEACHING EXAMPLE

2,000 kg steel × 1.9 kgCO2e/kg = 3,800 kgCO2e; compare per identical functional unit.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Compare identical functional units Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR IMPACT = Σ ACTIVITY DATA × FACTOR

Life-cycle thinking: evidence artifact

ACTUAL ARTIFACT

Cradle-to-gate inventory

REQUIRED FIELDS

Life-cycle thinking boundary; reporting period; Impact = Σ activity data × factor; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Burden shifted to supplier stage The artifact supports only the stated boundary and period.

ANCHOR CRADLE-TO-GATE INVENTORY

Life-cycle thinking: failure diagnostic

FAILURE

Burden shifted to supplier stage

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Compare identical functional units

ANCHOR BURDEN SHIFTED TO SUPPLIER STAGE

Life-cycle thinking: executive decision gate

REQUIRED INPUT

Cradle-to-gate inventory

MEASUREMENT

Impact = Σ activity data × factor Worked basis: 2,000 kg steel × 1.9 kgCO2e/kg = 3,800 kgCO2e; compare per identical functional unit.

TOPIC-SPECIFIC THRESHOLD

Compare identical functional units

RETAINED RECORD

Life-cycle thinking decision record: source, calculation, reviewer, conditions and next review.

ANCHOR COMPARE IDENTICAL FUNCTIONAL UNITS

Climate physical risk: operating model

INPUT

Asset risk heatmap

TRANSFORM

Hazard–exposure–vulnerability model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR HAZARD–EXPOSURE–VULNERABILITY MODEL

Climate physical risk: worked calculation

FORMULA / DECISION LOGIC

Risk = hazard × exposure × vulnerability

SYNTHETIC TEACHING EXAMPLE

Hazard 5 × exposure 4 × vulnerability 3 = 60/125; escalate scores ≥50.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Escalate high-consequence sites Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR RISK = HAZARD × EXPOSURE × VULNERABILITY

Climate physical risk: evidence artifact

ACTUAL ARTIFACT

Asset risk heatmap

REQUIRED FIELDS

Climate physical risk boundary; reporting period; Risk = hazard × exposure × vulnerability; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Corporate average hides critical asset The artifact supports only the stated boundary and period.

ANCHOR ASSET RISK HEATMAP

Climate physical risk: failure diagnostic

FAILURE

Corporate average hides critical asset

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Escalate high-consequence sites

ANCHOR CORPORATE AVERAGE HIDES CRITICAL ASSET

Climate physical risk: executive decision gate

REQUIRED INPUT

Asset risk heatmap

MEASUREMENT

Risk = hazard × exposure × vulnerability Worked basis: Hazard 5 × exposure 4 × vulnerability 3 = 60/125; escalate scores ≥50.

TOPIC-SPECIFIC THRESHOLD

Escalate high-consequence sites

RETAINED RECORD

Climate physical risk decision record: source, calculation, reviewer, conditions and next review.

ANCHOR ESCALATE HIGH-CONSEQUENCE SITES

Technology readiness: operating model

INPUT

Pilot acceptance record

TRANSFORM

Stage-gate evidence model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR STAGE-GATE EVIDENCE MODEL

Technology readiness: worked calculation

FORMULA / DECISION LOGIC

Readiness gap = required TRL − current TRL

SYNTHETIC TEACHING EXAMPLE

Required TRL 8 − current TRL 6 = 2-stage gap; require pilot gate before endorsement.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Endorse conditionally with test gate Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR READINESS GAP = REQUIRED TRL − CURRENT TRL

Technology readiness: evidence artifact

ACTUAL ARTIFACT

Pilot acceptance record

REQUIRED FIELDS

Technology readiness boundary; reporting period; Readiness gap = required TRL − current TRL; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Vendor claim replaces test evidence The artifact supports only the stated boundary and period.

ANCHOR PILOT ACCEPTANCE RECORD

Technology readiness: failure diagnostic

FAILURE

Vendor claim replaces test evidence

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Endorse conditionally with test gate

ANCHOR VENDOR CLAIM REPLACES TEST EVIDENCE

Technology readiness: executive decision gate

REQUIRED INPUT

Pilot acceptance record

MEASUREMENT

Readiness gap = required TRL − current TRL
Worked basis: Required TRL 8 − current TRL 6 = 2-stage gap; require pilot gate before endorsement.

TOPIC-SPECIFIC THRESHOLD

Endorse conditionally with test gate

RETAINED RECORD

Technology readiness decision record: source, calculation, reviewer, conditions and next review.

ANCHOR ENDORSE CONDITIONALLY WITH TEST GATE

Engineering endorsement: operating model

INPUT

Signed engineering memo

TRANSFORM

Decision-rights workflow

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR DECISION-RIGHTS WORKFLOW

Engineering endorsement: worked calculation

FORMULA / DECISION LOGIC

Confidence = verified fields ÷ required fields

SYNTHETIC TEACHING EXAMPLE

5 verified fields ÷ 6 required = 83%; hold below 100% when compliance evidence is missing.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Endorse with conditions, owner and review date Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR CONFIDENCE = VERIFIED FIELDS ÷ REQUIRED FIELDS

Engineering endorsement: evidence artifact

ACTUAL ARTIFACT

Signed engineering memo

REQUIRED FIELDS

Engineering endorsement boundary; reporting period; Confidence = verified fields ÷ required fields; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

No residual-risk owner The artifact supports only the stated boundary and period.

ANCHOR SIGNED ENGINEERING MEMO

Engineering endorsement: failure diagnostic

FAILURE

No residual-risk owner

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Endorse with conditions, owner and review date

ANCHOR NO RESIDUAL-RISK OWNER

Engineering endorsement: executive decision gate

REQUIRED INPUT

Signed engineering memo

MEASUREMENT

Confidence = verified fields ÷ required fields Worked basis: 5 verified fields ÷ 6 required = 83%; hold below 100% when compliance evidence is missing.

TOPIC-SPECIFIC THRESHOLD

Endorse with conditions, owner and review date

RETAINED RECORD

Engineering endorsement decision record: source, calculation, reviewer, conditions and next review.

ANCHOR ENDORSE WITH CONDITIONS, OWNER AND REVIEW DATE

Environmental economics: operating model

INPUT

Scenario cashflow table

TRANSFORM

Externality-to-cashflow map

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR EXTERNALITY-TO-CASHFLOW MAP

Environmental economics: worked calculation

FORMULA / DECISION LOGIC

NPV = Σ cash flow/(1+r)^t − investment

SYNTHETIC TEACHING EXAMPLE

PV benefits S$620k − PV costs S$500k = NPV S$120k; test at 8% discount rate.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Test base, upside and downside cases Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR NPV = Σ CASH FLOW/(1+R)^T − INVESTMENT

Environmental economics: evidence artifact

ACTUAL ARTIFACT

Scenario cashflow table

REQUIRED FIELDS

Environmental economics boundary; reporting period; NPV = Σ cash flow/(1+r)^t − investment; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Carbon price omitted The artifact supports only the stated boundary and period.

ANCHOR SCENARIO CASHFLOW TABLE

Environmental economics: failure diagnostic

FAILURE

Carbon price omitted

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Test base, upside and downside cases

ANCHOR CARBON PRICE OMITTED

Environmental economics: executive decision gate

REQUIRED INPUT

Scenario cashflow table

MEASUREMENT

NPV = Σ cash flow/(1+r)^t − investment Worked basis: PV benefits S$620k − PV costs S$500k = NPV S$120k; test at 8% discount rate.

TOPIC-SPECIFIC THRESHOLD

Test base, upside and downside cases

RETAINED RECORD

Environmental economics decision record: source, calculation, reviewer, conditions and next review.

ANCHOR TEST BASE, UPSIDE AND DOWNSIDE CASES

Stakeholder mapping: operating model

INPUT

Stakeholder evidence register

TRANSFORM

Power–interest–impact matrix

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR POWER–INTEREST–IMPACT MATRIX

Stakeholder mapping: worked calculation

FORMULA / DECISION LOGIC

Priority = significance × influence × exposure

SYNTHETIC TEACHING EXAMPLE

Significance 5 × influence 4 × exposure 3 = 60/125; priority threshold ≥48.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Use documented scoring anchors Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR PRIORITY = SIGNIFICANCE × INFLUENCE × EXPOSURE

Stakeholder mapping: evidence artifact

ACTUAL ARTIFACT

Stakeholder evidence register

REQUIRED FIELDS

Stakeholder mapping boundary; reporting period; Priority = significance × influence × exposure; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Loudest voice treated as material The artifact supports only the stated boundary and period.

ANCHOR STAKEHOLDER EVIDENCE REGISTER

Stakeholder mapping: failure diagnostic

FAILURE

Loudest voice treated as material

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Use documented scoring anchors

ANCHOR LOUDEST VOICE TREATED AS MATERIAL

Stakeholder mapping: executive decision gate

REQUIRED INPUT

Stakeholder evidence register

MEASUREMENT

Priority = significance × influence × exposure
Worked basis: Significance 5 × influence 4 × exposure 3 = 60/125; priority threshold ≥48.

TOPIC-SPECIFIC THRESHOLD

Use documented scoring anchors

RETAINED RECORD

Stakeholder mapping decision record: source, calculation, reviewer, conditions and next review.

ANCHOR USE DOCUMENTED SCORING ANCHORS

Double materiality: operating model

INPUT

5×5 materiality matrix

TRANSFORM

Impact/financial lens crosswalk

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR IMPACT/FINANCIAL LENS CROSSWALK

Double materiality: worked calculation

FORMULA / DECISION LOGIC

Priority = max(impact score, financial score)

SYNTHETIC TEACHING EXAMPLE

Impact 4.6, financial 3.8 → priority=max(4.6,3.8)=4.6; include at threshold ≥4.0.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Report lens and threshold explicitly Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR PRIORITY = MAX(IMPACT SCORE, FINANCIAL SCORE)

Double materiality: evidence artifact

ACTUAL ARTIFACT

5×5 materiality matrix

REQUIRED FIELDS

Double materiality boundary; reporting period; Priority = max(impact score, financial score); preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Two lenses merged without explanation The artifact supports only the stated boundary and period.

ANCHOR 5×5 MATERIALITY MATRIX

Double materiality: failure diagnostic

FAILURE

Two lenses merged without explanation

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

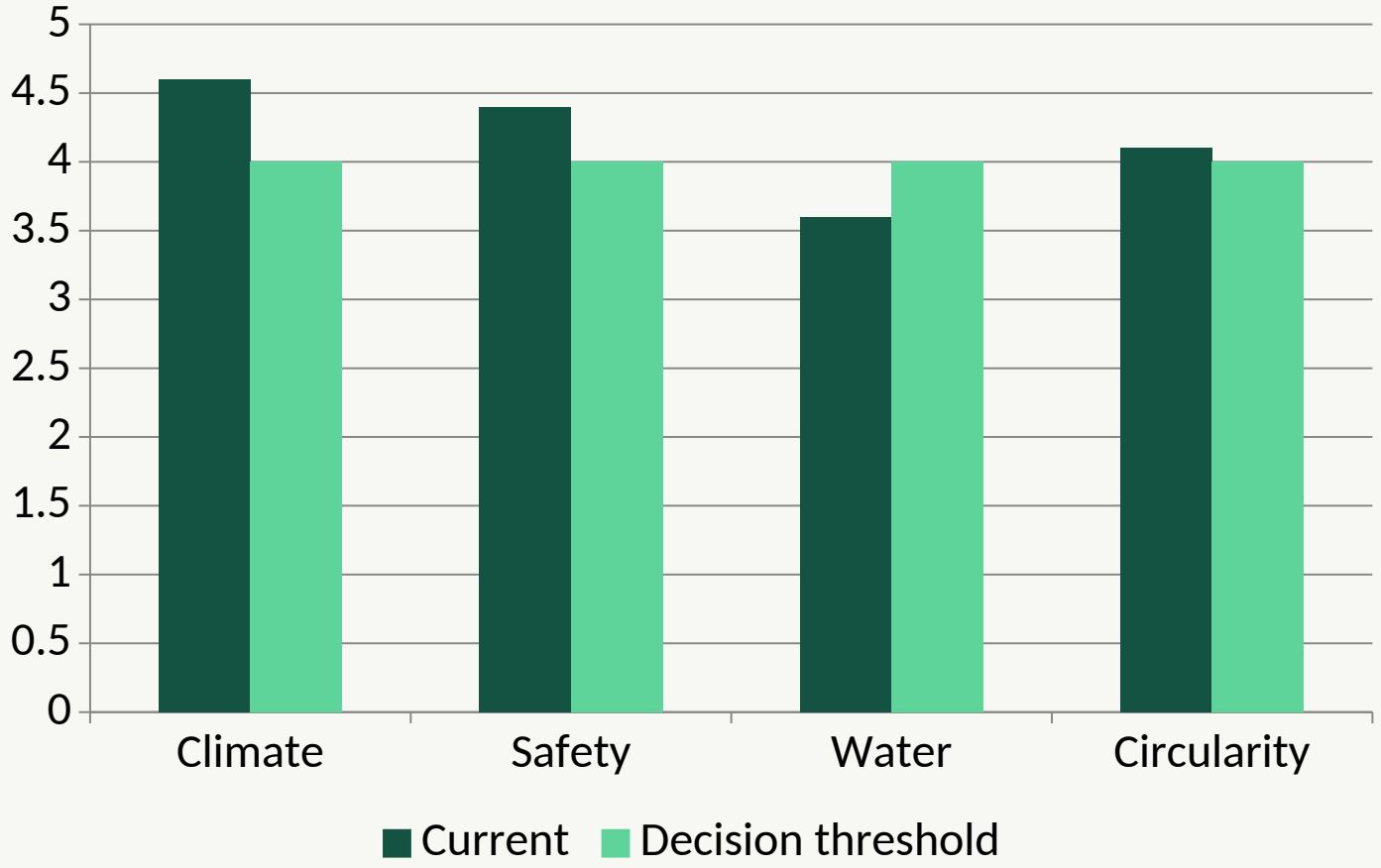
Reperform the calculation and trace the source.

REMEDIATE

Report lens and threshold explicitly

ANCHOR TWO LENSES MERGED WITHOUT EXPLANATION

Double materiality: executive decision gate



REQUIRED INPUT

5×5 materiality matrix

MEASUREMENT

Priority = max(impact score, financial score) Worked basis: Impact 4.6, financial 3.8 → priority=max(4.6,3.8)=4.6; include at threshold ≥4.0.

ANCHOR REPORT LENS AND THRESHOLD EXPLICITLY

ESG strategy: operating model

INPUT

Strategy-on-a-page

TRANSFORM

Issue-to-objective cascade

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR ISSUE-TO-OBJECTIVE CASCADE

ESG strategy: worked calculation

FORMULA / DECISION LOGIC

Gap = target − forecast

SYNTHETIC TEACHING EXAMPLE

2030 target 50% − forecast 38% = 12 percentage-point gap requiring funded initiatives.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Fund only initiatives tied to KPI Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR GAP = TARGET − FORECAST

ESG strategy: evidence artifact

ACTUAL ARTIFACT

Strategy-on-a-page

REQUIRED FIELDS

ESG strategy boundary; reporting period; Gap = target − forecast; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Initiative has no target The artifact supports only the stated boundary and period.

ANCHOR STRATEGY-ON-A-PAGE

ESG strategy: failure diagnostic

FAILURE

Initiative has no target

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Fund only initiatives tied to KPI

ANCHOR INITIATIVE HAS NO TARGET

LU2 retained source evidence: annotated decision use



SOURCE

Original slide 200 from “WSQ - Master Trainer Slides - Sustainability Reporting and Engineering Strategies for Senior Leaders - v3.pptx”.

INTERPRET

Use the source visual to identify stakeholder/market evidence, then connect it to a measurable strategic KPI.

CONTROL

Record boundary, method, owner, threshold and source citation before using the evidence.

Sustainability scorecard: operating model

INPUT

Owner/cadence/tolerance table

TRANSFORM

KPI definition sheet

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR KPI DEFINITION SHEET

Sustainability scorecard: worked calculation

FORMULA / DECISION LOGIC

Variance = actual − target

SYNTHETIC TEACHING EXAMPLE

Actual 64% − target 70% = −6 points; red when more than 5 points below target.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Change control every definition Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR VARIANCE = ACTUAL − TARGET

Sustainability scorecard: evidence artifact

ACTUAL ARTIFACT

Owner/cadence/tolerance table

REQUIRED FIELDS

Sustainability scorecard boundary; reporting period; Variance = actual − target; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Metric boundary changes mid-year The artifact supports only the stated boundary and period.

ANCHOR OWNER/CADENCE/TOLERANCE TABLE

Sustainability scorecard: failure diagnostic

FAILURE

Metric boundary changes mid-year

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Change control every definition

ANCHOR METRIC BOUNDARY CHANGES MID-YEAR

Sustainability scorecard: executive decision gate

REQUIRED INPUT

Owner/cadence/tolerance table

MEASUREMENT

Variance = actual − target Worked basis: Actual 64% − target 70% = −6 points; red when more than 5 points below target.

TOPIC-SPECIFIC THRESHOLD

Change control every definition

RETAINED RECORD

Sustainability scorecard decision record: source, calculation, reviewer, conditions and next review.

ANCHOR CHANGE CONTROL EVERY DEFINITION

Peer benchmarking: operating model

INPUT

Peer-normalisation table

TRANSFORM

Like-for-like comparison

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR LIKE-FOR-LIKE COMPARISON

Peer benchmarking: worked calculation

FORMULA / DECISION LOGIC

Indexed performance = company ÷ peer median

SYNTHETIC TEACHING EXAMPLE

Company 78 ÷ peer median 65 = 1.20 index; validate identical boundary before conclusion.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Reject non-comparable metrics Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR INDEXED PERFORMANCE = COMPANY ÷ PEER MEDIAN

Peer benchmarking: evidence artifact

ACTUAL ARTIFACT

Peer-normalisation table

REQUIRED FIELDS

Peer benchmarking boundary; reporting period; Indexed performance = company ÷ peer median; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Different boundary/period compared The artifact supports only the stated boundary and period.

ANCHOR PEER-NORMALISATION TABLE

Peer benchmarking: failure diagnostic

FAILURE

Different boundary/period compared

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Reject non-comparable metrics

ANCHOR DIFFERENT BOUNDARY/PERIOD COMPARED

Peer benchmarking: executive decision gate

REQUIRED INPUT

Peer-normalisation table

MEASUREMENT

Indexed performance = company ÷ peer median
Worked basis: Company 78 ÷ peer median 65 = 1.20 index; validate identical boundary before conclusion.

TOPIC-SPECIFIC THRESHOLD

Reject non-comparable metrics

RETAINED RECORD

Peer benchmarking decision record: source, calculation, reviewer, conditions and next review.

ANCHOR REJECT NON-COMPARABLE METRICS

ESG ratings: operating model

INPUT

Ratings divergence matrix

TRANSFORM

Rating-methodology trace

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR RATING-METHODOLOGY TRACE

ESG ratings: worked calculation

FORMULA / DECISION LOGIC

Sensitivity = score change per assumption

SYNTHETIC TEACHING EXAMPLE

Assumption change +1 causes score +0.35; sensitivity 0.35 points per assumption unit.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Use ratings as signals, not assurance Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR SENSITIVITY = SCORE CHANGE PER ASSUMPTION

ESG ratings: evidence artifact

ACTUAL ARTIFACT

Ratings divergence matrix

REQUIRED FIELDS

ESG ratings boundary; reporting period; Sensitivity = score change per assumption; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

One rating treated as ground truth The artifact supports only the stated boundary and period.

ANCHOR RATINGS DIVERGENCE MATRIX

ESG ratings: failure diagnostic

FAILURE

One rating treated as ground truth

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Use ratings as signals, not assurance

ANCHOR ONE RATING TREATED AS GROUND TRUTH

ESG ratings: executive decision gate

REQUIRED INPUT

Ratings divergence matrix

MEASUREMENT

Sensitivity = score change per assumption Worked basis: Assumption change +1 causes score +0.35; sensitivity 0.35 points per assumption unit.

TOPIC-SPECIFIC THRESHOLD

Use ratings as signals, not assurance

RETAINED RECORD

ESG ratings decision record: source, calculation, reviewer, conditions and next review.

ANCHOR USE RATINGS AS SIGNALS, NOT ASSURANCE

Risk treatment: operating model

INPUT

Risk/control register

TRANSFORM

Bow-tie control model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR BOW-TIE CONTROL MODEL

Risk treatment: worked calculation

FORMULA / DECISION LOGIC

Residual risk = inherent risk × (1 − control effectiveness)

SYNTHETIC TEACHING EXAMPLE

Inherent risk 20 × (1 − 70% control effectiveness) = residual risk 6; accept threshold ≤8.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Test design and operating effectiveness Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR RESIDUAL RISK = INHERENT RISK × (1 − CONTROL EFFECTIVENESS)

Risk treatment: evidence artifact

ACTUAL ARTIFACT

Risk/control register

REQUIRED FIELDS

Risk treatment boundary; reporting period; Residual risk = inherent risk × (1 − control effectiveness); preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Control exists only on paper The artifact supports only the stated boundary and period.

ANCHOR RISK/CONTROL REGISTER

Risk treatment: failure diagnostic

FAILURE

Control exists only on paper

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Test design and operating effectiveness

ANCHOR CONTROL EXISTS ONLY ON PAPER

Risk treatment: executive decision gate

REQUIRED INPUT

Risk/control register

MEASUREMENT

Residual risk = inherent risk × (1 − control effectiveness) Worked basis: Inherent risk 20 × (1 − 70% control effectiveness) = residual risk 6; accept threshold ≤8.

TOPIC-SPECIFIC THRESHOLD

Test design and operating effectiveness

RETAINED RECORD

Risk treatment decision record: source, calculation, reviewer, conditions and next review.

ANCHOR TEST DESIGN AND OPERATING EFFECTIVENESS

Market positioning: operating model

INPUT

Claims evidence library

TRANSFORM

Claim–proof–audience canvas

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR CLAIM–PROOF–AUDIENCE CANVAS

Market positioning: worked calculation

FORMULA / DECISION LOGIC

Proof coverage = supported claims ÷ total claims

SYNTHETIC TEACHING EXAMPLE

8 supported claims ÷ 10 total claims = 80% proof coverage; remove two unsupported claims.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Legal and evidence gate before release Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR PROOF COVERAGE = SUPPORTED CLAIMS ÷ TOTAL CLAIMS

Market positioning: evidence artifact

ACTUAL ARTIFACT

Claims evidence library

REQUIRED FIELDS

Market positioning boundary; reporting period; Proof coverage = supported claims ÷ total claims; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Aspirational claim stated as achieved The artifact supports only the stated boundary and period.

ANCHOR CLAIMS EVIDENCE LIBRARY

Market positioning: failure diagnostic

FAILURE

Aspirational claim stated as achieved

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Legal and evidence gate before release

ANCHOR ASPIRATIONAL CLAIM STATED AS ACHIEVED

Market positioning: executive decision gate

REQUIRED INPUT

Claims evidence library

MEASUREMENT

Proof coverage = supported claims ÷ total claims
Worked basis: 8 supported claims ÷ 10 total claims = 80% proof coverage; remove two unsupported claims.

TOPIC-SPECIFIC THRESHOLD

Legal and evidence gate before release

RETAINED RECORD

Market positioning decision record: source, calculation, reviewer, conditions and next review.

ANCHOR LEGAL AND EVIDENCE GATE BEFORE RELEASE

Portfolio governance: operating model

INPUT

Portfolio bubble chart

TRANSFORM

Initiative stage-gate

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR INITIATIVE STAGE-GATE

Portfolio governance: worked calculation

FORMULA / DECISION LOGIC

Portfolio score = 40% impact + 35% value + 25% feasibility

SYNTHETIC TEACHING EXAMPLE

0.40×impact 4.5 + 0.35×value 4.0 + 0.25×feasibility 3.0 = 3.95/5.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Prioritise scores ≥3.5/5, then test portfolio balance and dependencies Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR PORTFOLIO SCORE = 40% IMPACT + 35% VALUE + 25% FEASIBILITY

Portfolio governance: evidence artifact

ACTUAL ARTIFACT

Portfolio bubble chart

REQUIRED FIELDS

Portfolio governance boundary; reporting period; Portfolio score = 40% impact + 35% value + 25% feasibility; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Easy projects crowd out material risks The artifact supports only the stated boundary and period.

ANCHOR PORTFOLIO BUBBLE CHART

Portfolio governance: failure diagnostic

FAILURE

Easy projects crowd out material risks

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Prioritise scores ≥3.5/5, then test portfolio balance and dependencies

ANCHOR EASY PROJECTS CROWD OUT MATERIAL RISKS

Portfolio governance: executive decision gate

REQUIRED INPUT

Portfolio bubble chart

MEASUREMENT

Portfolio score = 40% impact + 35% value + 25% feasibility Worked basis: 0.40×impact 4.5 + 0.35×value 4.0 + 0.25×feasibility 3.0 = 3.95/5.

TOPIC-SPECIFIC THRESHOLD

Prioritise scores ≥3.5/5, then test portfolio balance and dependencies

RETAINED RECORD

Portfolio governance decision record: source, calculation, reviewer, conditions and next review.

ANCHOR PRIORITISE SCORES ≥3.5/5, THEN TEST PORTFOLIO BALANCE AND DEPENDENCIES

IFRS S1 architecture: operating model

INPUT

IFRS S1 crosswalk

TRANSFORM

Governance–strategy–risk–metrics model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR GOVERNANCE–STRATEGY–RISK–METRICS MODEL

IFRS S1 architecture: worked calculation

FORMULA / DECISION LOGIC

Coverage = disclosed requirements ÷ applicable requirements

SYNTHETIC TEACHING EXAMPLE

31 evidenced requirements ÷ 36 applicable = 86% coverage; close five gaps before issue.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Trace every disclosure to evidence owner Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR COVERAGE = DISCLOSED REQUIREMENTS ÷ APPLICABLE REQUIREMENTS

IFRS S1 architecture: evidence artifact

ACTUAL ARTIFACT

IFRS S1 crosswalk

REQUIRED FIELDS

IFRS S1 architecture boundary; reporting period; Coverage = disclosed requirements ÷ applicable requirements; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Boilerplate without entity facts The artifact supports only the stated boundary and period.

ANCHOR IFRS S1 CROSSWALK

IFRS S1 architecture: failure diagnostic

FAILURE

Boilerplate without entity facts

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Trace every disclosure to evidence owner

ANCHOR BOILERPLATE WITHOUT ENTITY FACTS

IFRS S1 architecture: executive decision gate

REQUIRED INPUT

IFRS S1 crosswalk

MEASUREMENT

Coverage = disclosed requirements ÷ applicable requirements Worked basis: 31 evidenced requirements ÷ 36 applicable = 86% coverage; close five gaps before issue.

TOPIC-SPECIFIC THRESHOLD

Trace every disclosure to evidence owner

RETAINED RECORD

IFRS S1 architecture decision record: source, calculation, reviewer, conditions and next review.

ANCHOR TRACE EVERY DISCLOSURE TO EVIDENCE OWNER

IFRS S2 climate: operating model

INPUT

Climate metrics register

TRANSFORM

Climate disclosure chain

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR CLIMATE DISCLOSURE CHAIN

IFRS S2 climate: worked calculation

FORMULA / DECISION LOGIC

Financed/operational emissions = activity × factor

SYNTHETIC TEACHING EXAMPLE

1,250,000 kWh × 0.408 kgCO2e/kWh = 510 tCO2e; record factor source/year.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Label assumptions and uncertainty Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR FINANCED/OPERATIONAL EMISSIONS = ACTIVITY × FACTOR

IFRS S2 climate: evidence artifact

ACTUAL ARTIFACT

Climate metrics register

REQUIRED FIELDS

IFRS S2 climate boundary; reporting period; Financed/operational emissions = activity × factor; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Scenario result presented as forecast The artifact supports only the stated boundary and period.

ANCHOR CLIMATE METRICS REGISTER

IFRS S2 climate: failure diagnostic

FAILURE

Scenario result presented as forecast

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Label assumptions and uncertainty

ANCHOR SCENARIO RESULT PRESENTED AS FORECAST

IFRS S2 climate: executive decision gate

REQUIRED INPUT

Climate metrics register

MEASUREMENT

Financed/operational emissions = activity × factor
Worked basis: 1,250,000 kWh × 0.408 kgCO2e/kWh = 510 tCO2e; record factor source/year.

TOPIC-SPECIFIC THRESHOLD

Label assumptions and uncertainty

RETAINED RECORD

IFRS S2 climate decision record: source, calculation, reviewer, conditions and next review.

ANCHOR LABEL ASSUMPTIONS AND UNCERTAINTY

GRI reporting: operating model

INPUT

GRI content index

TRANSFORM

Material-topic reporting flow

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR MATERIAL-TOPIC REPORTING FLOW

GRI reporting: worked calculation

FORMULA / DECISION LOGIC

Management effectiveness = outcome ÷ target

SYNTHETIC TEACHING EXAMPLE

Outcome 72% ÷ target 80% = 90% effectiveness; explain 8-point outcome gap.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Document determination and omission Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR MANAGEMENT EFFECTIVENESS = OUTCOME ÷ TARGET

GRI reporting: evidence artifact

ACTUAL ARTIFACT

GRI content index

REQUIRED FIELDS

GRI reporting boundary; reporting period; Management effectiveness = outcome ÷ target; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Topic omitted without rationale The artifact supports only the stated boundary and period.

ANCHOR GRI CONTENT INDEX

GRI reporting: failure diagnostic

FAILURE

Topic omitted without rationale

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Document determination and omission

ANCHOR TOPIC OMITTED WITHOUT RATIONALE

GRI reporting: executive decision gate

REQUIRED INPUT

GRI content index

MEASUREMENT

Management effectiveness = outcome ÷ target
Worked basis: Outcome 72% ÷ target 80% = 90% effectiveness; explain 8-point outcome gap.

TOPIC-SPECIFIC THRESHOLD

Document determination and omission

RETAINED RECORD

GRI reporting decision record: source, calculation, reviewer, conditions and next review.

ANCHOR DOCUMENT DETERMINATION AND OMISSION

SGX reporting: operating model

INPUT

SGX compliance checklist

TRANSFORM

Board-to-disclosure accountability

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR BOARD-TO-DISCLOSURE ACCOUNTABILITY

SGX reporting: worked calculation

FORMULA / DECISION LOGIC

Readiness = completed controls ÷ required controls

SYNTHETIC TEACHING EXAMPLE

18 completed controls ÷ 22 required = 81.8% readiness; four controls remain open.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Verify live rulebook each delivery Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR READINESS = COMPLETED CONTROLS ÷ REQUIRED CONTROLS

SGX reporting: evidence artifact

ACTUAL ARTIFACT

SGX compliance checklist

REQUIRED FIELDS

SGX reporting boundary; reporting period; Readiness = completed controls ÷ required controls; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Old roadmap date reused The artifact supports only the stated boundary and period.

ANCHOR SGX COMPLIANCE CHECKLIST

SGX reporting: failure diagnostic

FAILURE

Old roadmap date reused

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Verify live rulebook each delivery

ANCHOR OLD ROADMAP DATE REUSED

LU3 retained source evidence: annotated decision use

3 Governance

Topic	Metric	Unit	Source	Description
Board Composition	Board independence	Percentage (%)	GRI 102-22, WEF core metrics	The number of independent board directors as a percentage of all directors.
	Women on the board	Percentage (%)	GRI 102-22, GRI 405-1, WEF core metrics	The number of female board directors as a percentage of all directors.
Management Diversity	Women in the management team	Percentage (%)	GRI 102-22, GRI 405-1, WEF core metrics, SASB 330	The number of female senior management as a percentage of senior management. Each organisation defines which employees are part of its senior management team.
Ethical Behaviour	Anti-corruption disclosures	Discussion and number of standards	GRI 205-1, GRI 205-2 and GRI 205-3	Disclosures based on GRI's anti-corruption standards of 205-1, 205-2 and 205-3.
	Anti-corruption training for employees	Number and Percentage (%)	GRI 205-2, WEF core metrics	Number and percentage of employees that received anti-corruption training during reporting period.
Certifications	List of relevant certifications	List	Commonly reported metric by SGX issuers	List all sustainability or ESG-related certification (e.g. ISO 45000 family, BCA Green Building, LEED, ENERGY STAR). Each organisation defines which certifications are relevant to be reported.
Alignment with Frameworks	Alignment with frameworks and disclosure practices	GRI/ TCFD/ SASB/ SDGs/ others	SGX-ST Listing Rules (Mainboard) 711A and 711B, Practice Note 7.6; SGX-ST Listing Rules (Catalist) 711A and 711B, Practice Note 7F	The issuer needs to give priority to using globally-recognised frameworks and disclosure practices to guide its sustainability reporting. Where the issuer is applying a portion of a particular framework, the issuer should provide a general description of the extent of the issuer's application of the framework.
Assurance	Assurance of sustainability report	Internal/External/None	SGX-ST Listing Rules (Mainboard) 711A and 711B, Practice Note 7.6; SGX-ST Listing Rules (Catalist) 711A and 711B, Practice Note 7F	Disclose whether sustainability report has undertaken: (a) external independent assurance, (b) internal assurance or (c) no assurance. Provide scope of assurance if organisation has undertaken external or internal assurance.

SOURCE

Original slide 337 from "WSQ - Master Trainer Slides - Sustainability Reporting and Engineering Strategies for Senior Leaders - v3.pptx".

INTERPRET

Trace the reporting artifact to requirement, metric owner, calculation and retained assurance evidence.

CONTROL

Record boundary, method, owner, threshold and source citation before using the evidence.

Singapore roadmap: operating model

INPUT

Applicability decision log

TRANSFORM

Entity/year applicability decision table

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR ENTITY/YEAR APPLICABILITY DECISION TABLE

Singapore roadmap: worked calculation

FORMULA / DECISION LOGIC

Decision fields: listing/entity category; FY; framework; assurance phase; source date; outcome

SYNTHETIC TEACHING EXAMPLE

Example decision table: STI constituent; FY2027; ISSB-aligned climate disclosure; assurance/phasing status checked against ACRA 25 Aug 2025 roadmap → applicability review required before conclusion.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Separate law, rule and consultation; verify ACRA/SGX source and effective date Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR DECISION FIELDS: LISTING/ENTITY CATEGORY; FY; FRAMEWORK; ASSURANCE PHASE; SOURCE DATE; OUTCOME

Singapore roadmap: evidence artifact

ACTUAL ARTIFACT

Applicability decision log

REQUIRED FIELDS

Singapore roadmap boundary; reporting period; Decision fields: listing/entity category; FY; framework; assurance phase; source date; outcome; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Draft SDS treated as enacted The artifact supports only the stated boundary and period.

ANCHOR APPLICABILITY DECISION LOG

Singapore roadmap: failure diagnostic

FAILURE

Draft SDS treated as enacted

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Separate law, rule and consultation; verify ACRA/SGX source and effective date

ANCHOR DRAFT SDS TREATED AS ENACTED

Singapore roadmap: executive decision gate

REQUIRED INPUT

Applicability decision log

MEASUREMENT

Decision fields: listing/entity category; FY; framework; assurance phase; source date; outcome
Worked basis: Example decision table: STI constituent; FY2027; ISSB-aligned climate disclosure; assurance/phasing status checked against ACRA 25 Aug 2025 roadmap → applicability

TOPIC-SPECIFIC THRESHOLD

Separate law, rule and consultation; verify ACRA/SGX source and effective date

RETAINED RECORD

Singapore roadmap decision record: source, calculation, reviewer, conditions and next review.

ANCHOR SEPARATE LAW, RULE AND CONSULTATION; VERIFY ACRA/SGX SOURCE AND EFFECTIVE DATE

Data collection: operating model

INPUT

Data lineage map

TRANSFORM

Source-to-report lineage

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR SOURCE-TO-REPORT LINEAGE

Data collection: worked calculation

FORMULA / DECISION LOGIC

Reconciliation variance = ledger − disclosure

SYNTHETIC TEACHING EXAMPLE

Disclosure ledger 12,480 t − source ledger 12,420 t = 60 t variance (0.48%); tolerance 0.25%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Retain immutable source and approvals Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR RECONCILIATION VARIANCE = LEDGER − DISCLOSURE

Data collection: evidence artifact

ACTUAL ARTIFACT

Data lineage map

REQUIRED FIELDS

Data collection boundary; reporting period; Reconciliation variance = ledger − disclosure; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Spreadsheet overwrite breaks trace The artifact supports only the stated boundary and period.

ANCHOR DATA LINEAGE MAP

Data collection: failure diagnostic

FAILURE

Spreadsheet overwrite breaks trace

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

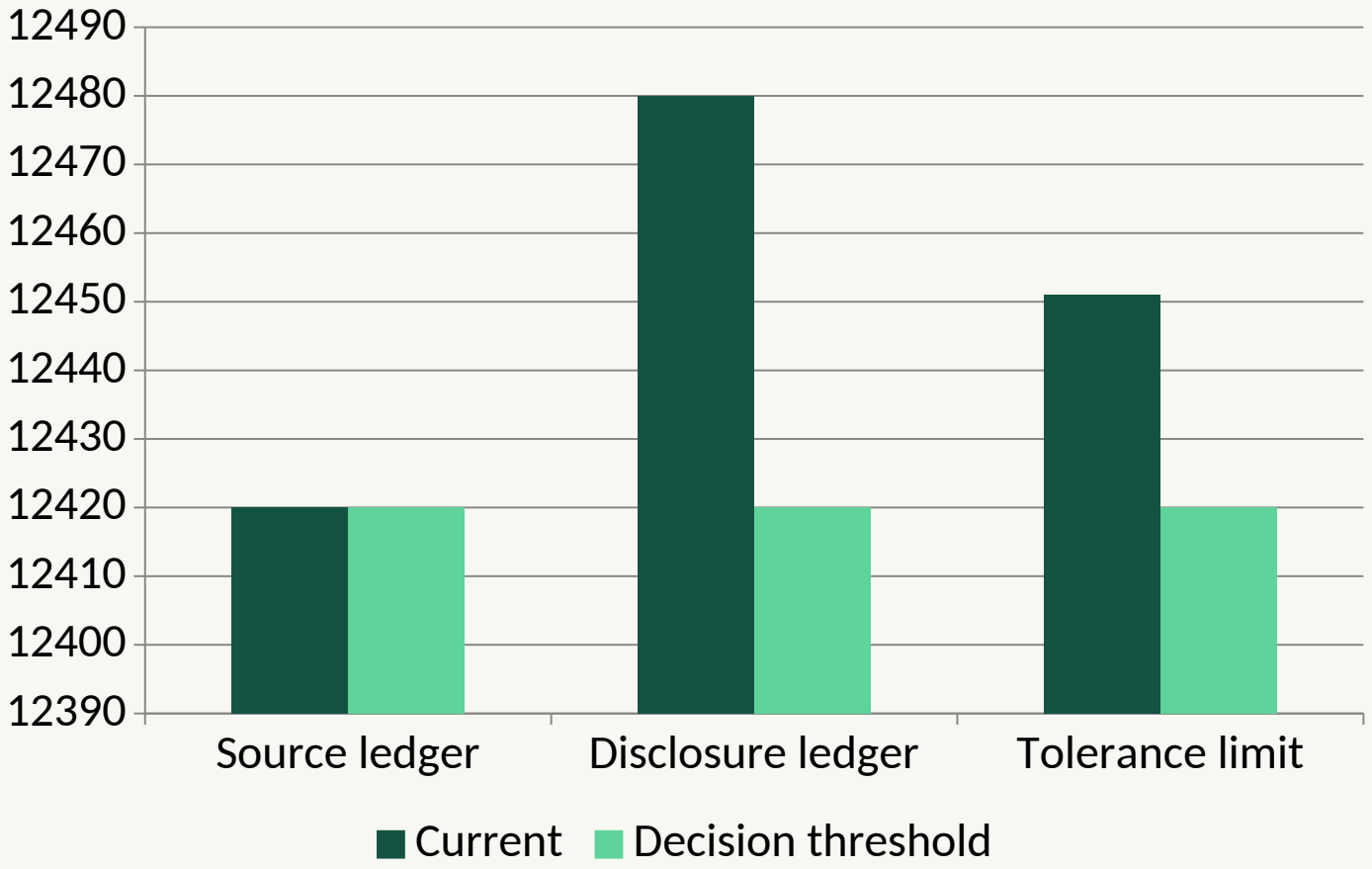
Reperform the calculation and trace the source.

REMEDIATE

Retain immutable source and approvals

ANCHOR SPREADSHEET OVERWRITE BREAKS TRACE

Data collection: executive decision gate



REQUIRED INPUT

Data lineage map

MEASUREMENT

Reconciliation variance = ledger − disclosure Worked basis: Disclosure ledger 12,480 t − source ledger 12,420 t = 60 t variance (0.48%); tolerance 0.25%.

ANCHOR RETAIN IMMUTABLE SOURCE AND APPROVALS

Metric controls: operating model

INPUT

Control design sheet

TRANSFORM

Prevent–detect–correct model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR PREVENT–DETECT–CORRECT MODEL

Metric controls: worked calculation

FORMULA / DECISION LOGIC

Control risk = likelihood × disclosure impact

SYNTHETIC TEACHING EXAMPLE

Likelihood 4 × disclosure impact 5 = risk 20; mandatory escalation at ≥15.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Segregate preparation and approval Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR CONTROL RISK = LIKELIHOOD × DISCLOSURE IMPACT

Metric controls: evidence artifact

ACTUAL ARTIFACT

Control design sheet

REQUIRED FIELDS

Metric controls boundary; reporting period; Control risk = likelihood × disclosure impact; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Owner and reviewer are same The artifact supports only the stated boundary and period.

ANCHOR CONTROL DESIGN SHEET

Metric controls: failure diagnostic

FAILURE

Owner and reviewer are same

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Segregate preparation and approval

ANCHOR OWNER AND REVIEWER ARE SAME

Metric controls: executive decision gate

REQUIRED INPUT

Control design sheet

MEASUREMENT

Control risk = likelihood × disclosure impact Worked basis: Likelihood 4 × disclosure impact 5 = risk 20; mandatory escalation at ≥15.

TOPIC-SPECIFIC THRESHOLD

Segregate preparation and approval

RETAINED RECORD

Metric controls decision record: source, calculation, reviewer, conditions and next review.

ANCHOR SEGREGATE PREPARATION AND APPROVAL

Assurance readiness: operating model

INPUT

Assurance PBC list

TRANSFORM

Evidence sufficiency matrix

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR EVIDENCE SUFFICIENCY MATRIX

Assurance readiness: worked calculation

FORMULA / DECISION LOGIC

Exception rate = failed samples ÷ tested samples

SYNTHETIC TEACHING EXAMPLE

3 failed samples ÷ 40 tested = 7.5% exception rate; remediate above 5%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Test high-risk assertions first Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR EXCEPTION RATE = FAILED SAMPLES ÷ TESTED SAMPLES

Assurance readiness: evidence artifact

ACTUAL ARTIFACT

Assurance PBC list

REQUIRED FIELDS

Assurance readiness boundary; reporting period; Exception rate = failed samples ÷ tested samples; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Narrative claim lacks retained record The artifact supports only the stated boundary and period.

ANCHOR ASSURANCE PBC LIST

Assurance readiness: failure diagnostic

FAILURE

Narrative claim lacks retained record

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Test high-risk assertions first

ANCHOR NARRATIVE CLAIM LACKS RETAINED RECORD

Assurance readiness: executive decision gate

REQUIRED INPUT

Assurance PBC list

MEASUREMENT

Exception rate = failed samples ÷ tested samples
Worked basis: 3 failed samples ÷ 40 tested = 7.5% exception rate; remediate above 5%.

TOPIC-SPECIFIC THRESHOLD

Test high-risk assertions first

RETAINED RECORD

Assurance readiness decision record: source, calculation, reviewer, conditions and next review.

ANCHOR TEST HIGH-RISK ASSERTIONS FIRST

Disclosure drafting: operating model

INPUT

Disclosure annotation sheet

TRANSFORM

Fact–judgement–estimate labels

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR FACT–JUDGEMENT–ESTIMATE LABELS

Disclosure drafting: worked calculation

FORMULA / DECISION LOGIC

Estimate range = high − low

SYNTHETIC TEACHING EXAMPLE

High estimate 1,200 − low 900 = 300 tCO2e range; disclose method and sensitivity.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

State method, range and sensitivity Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR ESTIMATE RANGE = HIGH − LOW

Disclosure drafting: evidence artifact

ACTUAL ARTIFACT

Disclosure annotation sheet

REQUIRED FIELDS

Disclosure drafting boundary; reporting period; Estimate range = high − low; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Estimate presented as fact The artifact supports only the stated boundary and period.

ANCHOR DISCLOSURE ANNOTATION SHEET

Disclosure drafting: failure diagnostic

FAILURE

Estimate presented as fact

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

State method, range and sensitivity

ANCHOR ESTIMATE PRESENTED AS FACT

Disclosure drafting: executive decision gate

REQUIRED INPUT

Disclosure annotation sheet

MEASUREMENT

Estimate range = high − low Worked basis: High estimate 1,200 − low 900 = 300 tCO2e range; disclose method and sensitivity.

TOPIC-SPECIFIC THRESHOLD

State method, range and sensitivity

RETAINED RECORD

Disclosure drafting decision record: source, calculation, reviewer, conditions and next review.

ANCHOR STATE METHOD, RANGE AND SENSITIVITY

Compliance governance: operating model

INPUT

Compliance remediation log

TRANSFORM

Obligation-to-control crosswalk

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR OBLIGATION-TO-CONTROL CROSSWALK

Compliance governance: worked calculation

FORMULA / DECISION LOGIC

Closure rate = closed gaps ÷ total gaps

SYNTHETIC TEACHING EXAMPLE

17 closed gaps ÷ 20 total = 85% closure; three effective-date gaps remain.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Assign legal owner and trigger review Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR CLOSURE RATE = CLOSED GAPS ÷ TOTAL GAPS

Compliance governance: evidence artifact

ACTUAL ARTIFACT

Compliance remediation log

REQUIRED FIELDS

Compliance governance boundary; reporting period; Closure rate = closed gaps ÷ total gaps; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

No effective-date monitoring The artifact supports only the stated boundary and period.

ANCHOR COMPLIANCE REMEDIATION LOG

Compliance governance: failure diagnostic

FAILURE

No effective-date monitoring

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Assign legal owner and trigger review

ANCHOR NO EFFECTIVE-DATE MONITORING

Compliance governance: executive decision gate

REQUIRED INPUT

Compliance remediation log

MEASUREMENT

Closure rate = closed gaps ÷ total gaps Worked basis: 17 closed gaps ÷ 20 total = 85% closure; three effective-date gaps remain.

TOPIC-SPECIFIC THRESHOLD

Assign legal owner and trigger review

RETAINED RECORD

Compliance governance decision record: source, calculation, reviewer, conditions and next review.

ANCHOR ASSIGN LEGAL OWNER AND TRIGGER REVIEW

Closed-loop manufacturing: physical evidence map



1 RECOVER

Heat, water and material streams return to defined process points.

2 MEASURE

Meters and mass-balance records establish the baseline and avoided impact.

3 GOVERN

Approve only with boundary, owner, threshold and retained evidence.

Circular economy: worked calculation

FORMULA / DECISION LOGIC

Circular input rate = circular input ÷ total input

SYNTHETIC TEACHING EXAMPLE

360 t circular input ÷ 1,200 t total input = 30%; target 40% → 120 t gap.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Prioritise refuse, reduce, reuse, repair Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR CIRCULAR INPUT RATE = CIRCULAR INPUT ÷ TOTAL INPUT

Circular economy: evidence artifact

ACTUAL ARTIFACT

Material-flow loop

REQUIRED FIELDS

Circular economy boundary; reporting period; Circular input rate = circular input ÷ total input; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Recycling used as only strategy The artifact supports only the stated boundary and period.

ANCHOR MATERIAL-FLOW LOOP

Circular economy: failure diagnostic

FAILURE

Recycling used as only strategy

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Prioritise refuse, reduce, reuse, repair

ANCHOR RECYCLING USED AS ONLY STRATEGY

Circular economy: executive decision gate

REQUIRED INPUT

Material-flow loop

MEASUREMENT

Circular input rate = circular input ÷ total input
Worked basis: 360 t circular input ÷ 1,200 t total input = 30%; target 40% → 120 t gap.

TOPIC-SPECIFIC THRESHOLD

Prioritise refuse, reduce, reuse, repair

RETAINED RECORD

Circular economy decision record: source, calculation, reviewer, conditions and next review.

ANCHOR PRIORITISE REFUSE, REDUCE, REUSE, REPAIR

Material flow: operating model

INPUT

Sankey-style balance table

TRANSFORM

Mass-balance model

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR MASS-BALANCE MODEL

Material flow: worked calculation

FORMULA / DECISION LOGIC

Variance = input − product − recovery − waste

SYNTHETIC TEACHING EXAMPLE

1,000 − 620 − 280 − 80 = 20 t variance (2%); investigate above 1%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Investigate variance above tolerance Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR VARIANCE = INPUT − PRODUCT − RECOVERY − WASTE

Material flow: evidence artifact

ACTUAL ARTIFACT

Sankey-style balance table

REQUIRED FIELDS

Material flow boundary; reporting period; Variance = input − product − recovery − waste; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Unexplained losses ignored The artifact supports only the stated boundary and period.

ANCHOR SANKEY-STYLE BALANCE TABLE

Material flow: failure diagnostic

FAILURE

Unexplained losses ignored

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Investigate variance above tolerance

ANCHOR UNEXPLAINED LOSSES IGNORED

Material flow: executive decision gate

REQUIRED INPUT

Sankey-style balance table

MEASUREMENT

Variance = input − product − recovery − waste
Worked basis: 1,000 − 620 − 280 − 80 = 20 t variance (2%); investigate above 1%.

TOPIC-SPECIFIC THRESHOLD

Investigate variance above tolerance

RETAINED RECORD

Material flow decision record: source, calculation, reviewer, conditions and next review.

ANCHOR INVESTIGATE VARIANCE ABOVE TOLERANCE

Circular design: operating model

INPUT

Bill-of-material circularity sheet

TRANSFORM

Design-for-disassembly map

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR DESIGN-FOR-DISASSEMBLY MAP

Circular design: worked calculation

FORMULA / DECISION LOGIC

Recovery yield = recovered mass ÷ returned mass

SYNTHETIC TEACHING EXAMPLE

420 kg recovered ÷ 600 kg returned = 70% recovery yield; redesign target 85%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Verify system, not material claim alone Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR RECOVERY YIELD = RECOVERED MASS ÷ RETURNED MASS

Circular design: evidence artifact

ACTUAL ARTIFACT

Bill-of-material circularity sheet

REQUIRED FIELDS

Circular design boundary; reporting period; Recovery yield = recovered mass ÷ returned mass; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Recyclable but no collection route The artifact supports only the stated boundary and period.

ANCHOR BILL-OF-MATERIAL CIRCULARITY SHEET

Circular design: failure diagnostic

FAILURE

Recyclable but no collection route

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Verify system, not material claim alone

ANCHOR RECYCLABLE BUT NO COLLECTION ROUTE

Circular design: executive decision gate

REQUIRED INPUT

Bill-of-material circularity sheet

MEASUREMENT

Recovery yield = recovered mass ÷ returned mass
Worked basis: 420 kg recovered ÷ 600 kg returned = 70% recovery yield; redesign target 85%.

TOPIC-SPECIFIC THRESHOLD

Verify system, not material claim alone

RETAINED RECORD

Circular design decision record: source, calculation, reviewer, conditions and next review.

ANCHOR VERIFY SYSTEM, NOT MATERIAL CLAIM ALONE

Product take-back: operating model

INPUT

Take-back funnel

TRANSFORM

Reverse-logistics network

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR REVERSE-LOGISTICS NETWORK

Product take-back: worked calculation

FORMULA / DECISION LOGIC

Return rate = returned units ÷ sold units

SYNTHETIC TEACHING EXAMPLE

8,400 returned ÷ 60,000 sold = 14% return rate; target 20% → 3,600-unit gap.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Unique chain-of-custody record Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR RETURN RATE = RETURNED UNITS ÷ SOLD UNITS

Product take-back: evidence artifact

ACTUAL ARTIFACT

Take-back funnel

REQUIRED FIELDS

Product take-back boundary; reporting period; Return rate = returned units ÷ sold units; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Collected mass double-counted The artifact supports only the stated boundary and period.

ANCHOR TAKE-BACK FUNNEL

Product take-back: failure diagnostic

FAILURE

Collected mass double-counted

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

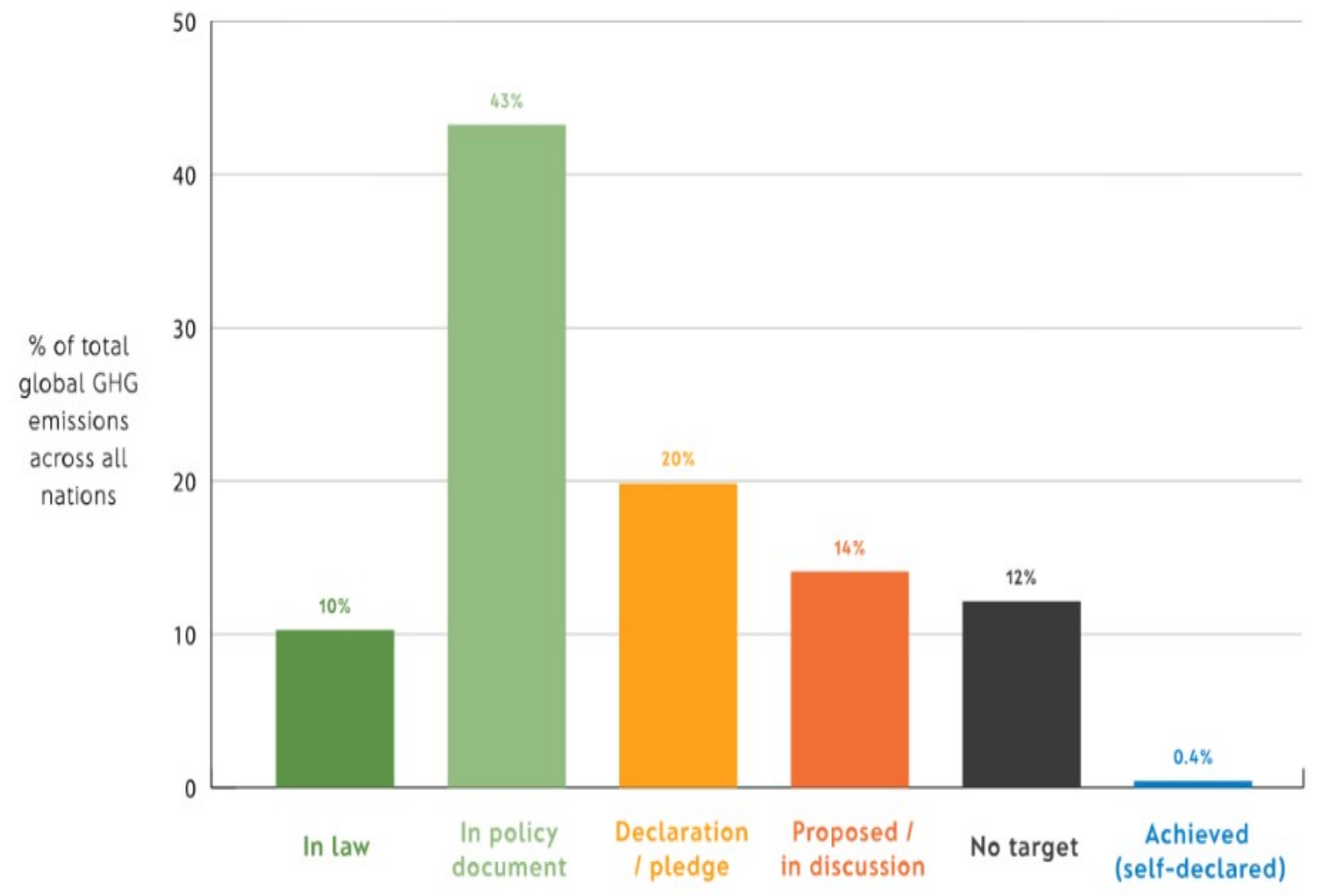
Reperform the calculation and trace the source.

REMEDIATE

Unique chain-of-custody record

ANCHOR COLLECTED MASS DOUBLE-COUNTED

LU4 retained source evidence: annotated decision use



SOURCE

Original slide 401 from “WSQ - Master Trainer Slides - Sustainability Reporting and Engineering Strategies for Senior Leaders - v3.pptx”.

INTERPRET

Use the original carbon/circularity evidence to define baseline, unit, reduction lever and verification method.

CONTROL

Record boundary, method, owner, threshold and source citation before using the evidence.

Sustainable procurement: operating model

INPUT

Weighted tender matrix

TRANSFORM

ISO 20400 control chain

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR ISO 20400 CONTROL CHAIN

Sustainable procurement: worked calculation

FORMULA / DECISION LOGIC

Supplier coverage = assessed spend ÷ addressable spend

SYNTHETIC TEACHING EXAMPLE

S$18m assessed spend ÷ S$24m addressable = 75% supplier coverage.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Embed criteria, evidence and remedy Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR SUPPLIER COVERAGE = ASSESSED SPEND ÷ ADDRESSABLE SPEND

Sustainable procurement: evidence artifact

ACTUAL ARTIFACT

Weighted tender matrix

REQUIRED FIELDS

Sustainable procurement boundary; reporting period; Supplier coverage = assessed spend ÷ addressable spend; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Policy without contract clause The artifact supports only the stated boundary and period.

ANCHOR WEIGHTED TENDER MATRIX

Sustainable procurement: failure diagnostic

FAILURE

Policy without contract clause

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Embed criteria, evidence and remedy

ANCHOR POLICY WITHOUT CONTRACT CLAUSE

Sustainable procurement: executive decision gate

REQUIRED INPUT

Weighted tender matrix

MEASUREMENT

Supplier coverage = assessed spend ÷ addressable spend Worked basis: S$18m assessed spend ÷ S$24m addressable = 75% supplier coverage.

TOPIC-SPECIFIC THRESHOLD

Embed criteria, evidence and remedy

RETAINED RECORD

Sustainable procurement decision record: source, calculation, reviewer, conditions and next review.

ANCHOR EMBED CRITERIA, EVIDENCE AND REMEDY

Supply-chain disclosure: operating model

INPUT

Supplier evidence register

TRANSFORM

Tiered traceability map

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR TIERED TRACEABILITY MAP

Supply-chain disclosure: worked calculation

FORMULA / DECISION LOGIC

Traceability = verified tier spend ÷ total spend

SYNTHETIC TEACHING EXAMPLE

S$9m verified tier spend ÷ S$15m total = 60% traceability; high-risk threshold 90%.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Risk-tier and validate samples Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR TRACEABILITY = VERIFIED TIER SPEND ÷ TOTAL SPEND

Supply-chain disclosure: evidence artifact

ACTUAL ARTIFACT

Supplier evidence register

REQUIRED FIELDS

Supply-chain disclosure boundary; reporting period; Traceability = verified tier spend ÷ total spend; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Survey answer treated as verified The artifact supports only the stated boundary and period.

ANCHOR SUPPLIER EVIDENCE REGISTER

Supply-chain disclosure: failure diagnostic

FAILURE

Survey answer treated as verified

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Risk-tier and validate samples

ANCHOR SURVEY ANSWER TREATED AS VERIFIED

Supply-chain disclosure: executive decision gate

REQUIRED INPUT

Supplier evidence register

MEASUREMENT

Traceability = verified tier spend ÷ total spend
Worked basis: S$9m verified tier spend ÷ S$15m total = 60% traceability; high-risk threshold 90%.

TOPIC-SPECIFIC THRESHOLD

Risk-tier and validate samples

RETAINED RECORD

Supply-chain disclosure decision record: source, calculation, reviewer, conditions and next review.

ANCHOR RISK-TIER AND VALIDATE SAMPLES

GHG inventory: operating model

INPUT

GHG inventory workbook

TRANSFORM

Scope 1–2–3 boundary map

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR SCOPE 1–2–3 BOUNDARY MAP

GHG inventory: worked calculation

FORMULA / DECISION LOGIC

tCO2e = activity data × emission factor

SYNTHETIC TEACHING EXAMPLE

500,000 L fuel × 2.68 kgCO2e/L = 1,340 tCO2e Scope 1.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Disclose method and factor source Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR TCO2E = ACTIVITY DATA × EMISSION FACTOR

GHG inventory: evidence artifact

ACTUAL ARTIFACT

GHG inventory workbook

REQUIRED FIELDS

GHG inventory boundary; reporting period; tCO2e = activity data × emission factor; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Market/location method mixed The artifact supports only the stated boundary and period.

ANCHOR GHG INVENTORY WORKBOOK

GHG inventory: failure diagnostic

FAILURE

Market/location method mixed

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Disclose method and factor source

ANCHOR MARKET/LOCATION METHOD MIXED

GHG inventory: executive decision gate

REQUIRED INPUT

GHG inventory workbook

MEASUREMENT

tCO2e = activity data × emission factor Worked basis: 500,000 L fuel × 2.68 kgCO2e/L = 1,340 tCO2e Scope 1.

TOPIC-SPECIFIC THRESHOLD

Disclose method and factor source

RETAINED RECORD

GHG inventory decision record: source, calculation, reviewer, conditions and next review.

ANCHOR DISCLOSE METHOD AND FACTOR SOURCE

Marginal abatement cost: operating model

INPUT

Abatement portfolio table

TRANSFORM

MAC curve

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR MAC CURVE

Marginal abatement cost: worked calculation

FORMULA / DECISION LOGIC

MAC = net annual cost ÷ annual tCO2e avoided

SYNTHETIC TEACHING EXAMPLE

−S$85,000 ÷ 420 tCO2e = −S$202/tCO2e; no-regret, then check dependency.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Sequence by MAC, dependency and risk Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR MAC = NET ANNUAL COST ÷ ANNUAL TCO2E AVOIDED

Marginal abatement cost: evidence artifact

ACTUAL ARTIFACT

Abatement portfolio table

REQUIRED FIELDS

Marginal abatement cost boundary; reporting period; MAC = net annual cost ÷ annual tCO2e avoided; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Initiatives added despite dependency The artifact supports only the stated boundary and period.

ANCHOR ABATEMENT PORTFOLIO TABLE

Marginal abatement cost: failure diagnostic

FAILURE

Initiatives added despite dependency

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

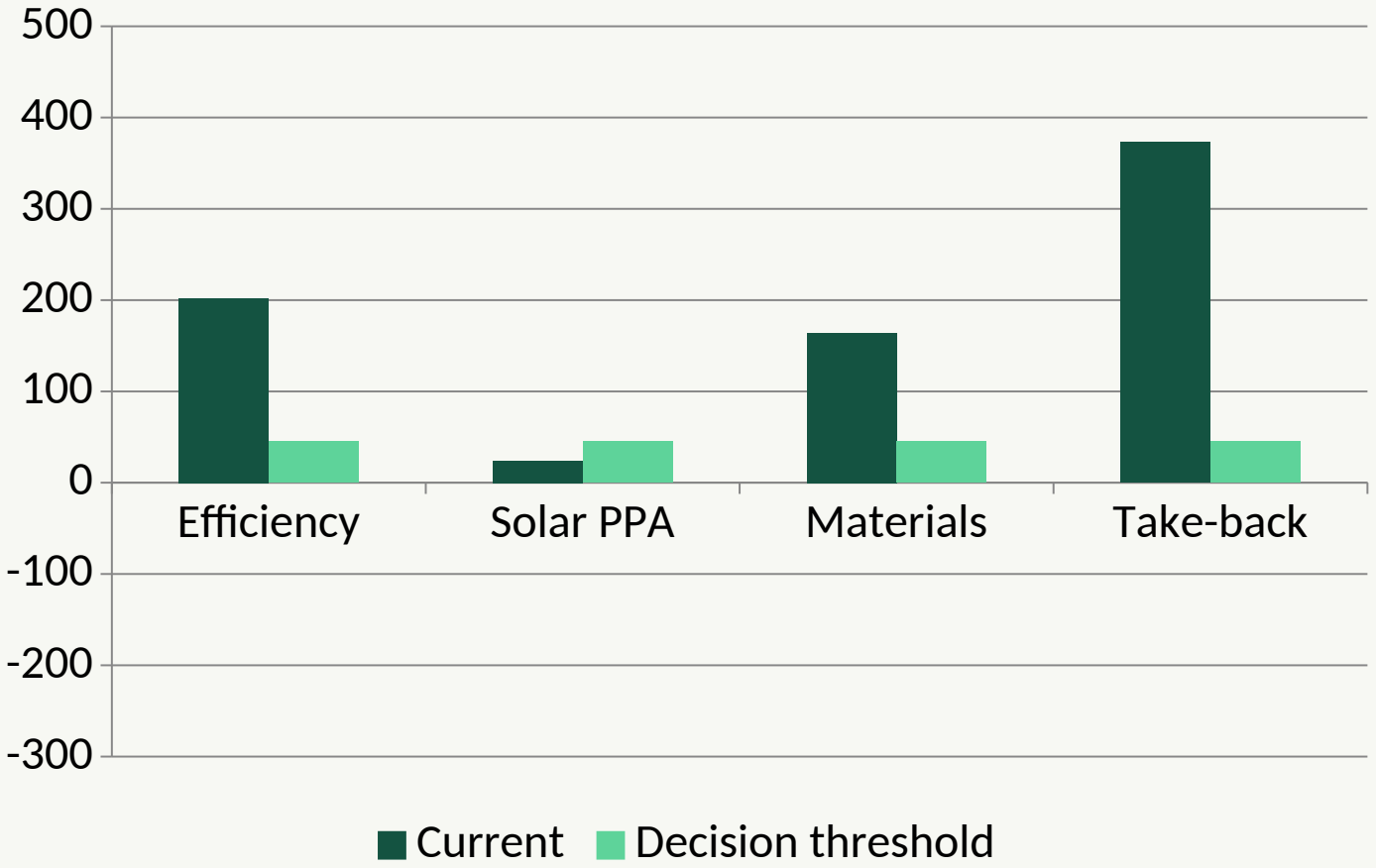
Reperform the calculation and trace the source.

REMEDIATE

Sequence by MAC, dependency and risk

ANCHOR INITIATIVES ADDED DESPITE DEPENDENCY

Marginal abatement cost: executive decision gate



REQUIRED INPUT

Abatement portfolio table

MEASUREMENT

MAC = net annual cost ÷ annual tCO2e avoided Worked basis: −S$85,000 ÷ 420 tCO2e = −S$202/tCO2e; no-regret, then check dependency.

ANCHOR SEQUENCE BY MAC, DEPENDENCY AND RISK

Carbon tax context: operating model

INPUT

Carbon-cost sensitivity table

TRANSFORM

Policy cost scenario

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR POLICY COST SCENARIO

Carbon tax context: worked calculation

FORMULA / DECISION LOGIC

Tax exposure = taxable tCO2e × S$45 (2026–27)

SYNTHETIC TEACHING EXAMPLE

12,000 taxable tCO2e × S$45 = S$540,000 exposure for 2026–27.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Include energy, capex and transition risk Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR TAX EXPOSURE = TAXABLE TCO2E × S$45 (2026–27)

Carbon tax context: evidence artifact

ACTUAL ARTIFACT

Carbon-cost sensitivity table

REQUIRED FIELDS

Carbon tax context boundary; reporting period; Tax exposure = taxable tCO2e × S$45 (2026–27); preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Tax alone used as business case The artifact supports only the stated boundary and period.

ANCHOR CARBON-COST SENSITIVITY TABLE

Carbon tax context: failure diagnostic

FAILURE

Tax alone used as business case

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Include energy, capex and transition risk

ANCHOR TAX ALONE USED AS BUSINESS CASE

Carbon tax context: executive decision gate

REQUIRED INPUT

Carbon-cost sensitivity table

MEASUREMENT

Tax exposure = taxable tCO2e × S$45 (2026–27)
Worked basis: 12,000 taxable tCO2e × S$45 = S$540,000 exposure for 2026–27.

TOPIC-SPECIFIC THRESHOLD

Include energy, capex and transition risk

RETAINED RECORD

Carbon tax context decision record: source, calculation, reviewer, conditions and next review.

ANCHOR INCLUDE ENERGY, CAPEX AND TRANSITION RISK

Sustainability portfolio: operating model

INPUT

Executive portfolio roadmap

TRANSFORM

Value–impact–feasibility matrix

OUTPUT

Decision-ready result with stated boundary.

GOVERN

Owner, reviewer, cadence and escalation.

ANCHOR VALUE–IMPACT–FEASIBILITY MATRIX

Sustainability portfolio: worked calculation

FORMULA / DECISION LOGIC

Portfolio score = 40% impact + 35% value + 25% feasibility

SYNTHETIC TEACHING EXAMPLE

0.40×impact 4.2 + 0.35×value 3.8 + 0.25×feasibility 3.6 = 3.91/5.

VALIDATION TEST

Reconcile source totals, confirm unit and period, then vary the most uncertain input ±10% where numerical.

INTERPRETATION

Define owner, indicator and verification Thresholds are course assumptions unless an official source is explicitly cited.

ANCHOR PORTFOLIO SCORE = 40% IMPACT + 35% VALUE + 25% FEASIBILITY

Sustainability portfolio: evidence artifact

ACTUAL ARTIFACT

Executive portfolio roadmap

REQUIRED FIELDS

Sustainability portfolio boundary; reporting period; Portfolio score = 40% impact + 35% value + 25% feasibility; preparer; reviewer; source reference.

CONTROL TEST

Select one reported value, reperform it from the retained source, and compare with the approved output.

ASSERTION LIMIT

Community effect left qualitative The artifact supports only the stated boundary and period.

ANCHOR EXECUTIVE PORTFOLIO ROADMAP

Sustainability portfolio: failure diagnostic

FAILURE

Community effect left qualitative

SIGNAL

Mismatch in boundary, method, period or ownership.

TEST

Reperform the calculation and trace the source.

REMEDIATE

Define owner, indicator and verification

ANCHOR COMMUNITY EFFECT LEFT QUALITATIVE

Sustainability portfolio: executive decision gate

REQUIRED INPUT

Executive portfolio roadmap

MEASUREMENT

Portfolio score = 40% impact + 35% value + 25% feasibility Worked basis: 0.40×impact 4.2 + 0.35×value 3.8 + 0.25×feasibility 3.6 = 3.91/5.

TOPIC-SPECIFIC THRESHOLD

Define owner, indicator and verification

RETAINED RECORD

Sustainability portfolio decision record: source, calculation, reviewer, conditions and next review.

ANCHOR DEFINE OWNER, INDICATOR AND VERIFICATION

Activity 1 debrief

ACTION

Reperform payback and weighted score.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR ACTIVITY 1 DEBRIEF

Activity 2 debrief

ACTION

Defend the materiality threshold and KPI tolerance.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR ACTIVITY 2 DEBRIEF

Activity 3 debrief

ACTION

Trace data to disclosure and control evidence.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR ACTIVITY 3 DEBRIEF

Activity 4 debrief

ACTION

Explain MAC, dependencies and verification.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR ACTIVITY 4 DEBRIEF

Assessment reminder

ACTION

Answer every item and save before upload.

EVIDENCE

Retain the relevant record.

OWNER

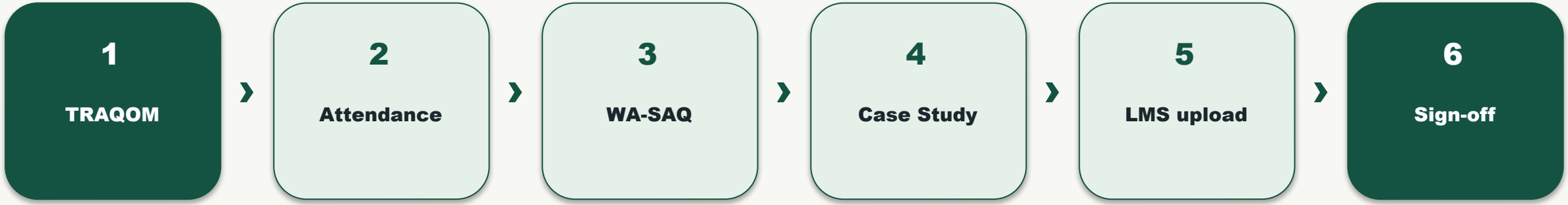
Name the accountable role.

NEXT GATE

Confirm completion before course close.

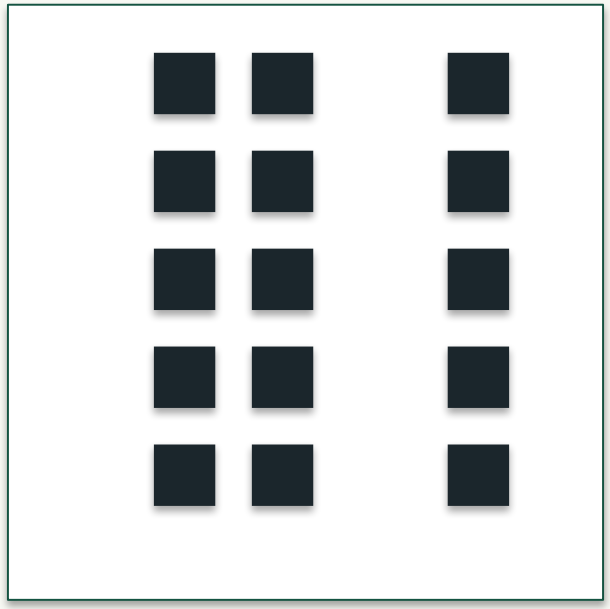
ANCHOR ASSESSMENT REMINDER

Assessment Flow



Sign-off record: grade • assessor • date • evidence location • appeal route

Assessment attendance confirmation



✓ Successful check-in

Verify learner name, course run and session. If the success state is absent, notify the trainer before proceeding.

Course feedback

ACTION

Submit TRAQOM and confirm receipt.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR COURSE FEEDBACK

Action plan

ACTION

Name one owner, metric, threshold and next review.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR ACTION PLAN

CLOSING

Thank You

ACTION

Engineering evidence → reporting confidence → strategic value.

EVIDENCE

Retain the relevant record.

OWNER

Name the accountable role.

NEXT GATE

Confirm completion before course close.

ANCHOR THANK YOU